

**Name:** Team Timbuktu

**Date:** 2/6/25

## ***Element C: Presentation and justification of solution design requirements***

### **1. Height (Highest Priority)**

The enrichment toy hangers must suspend toys 10 feet or more from the ground and have a vertical structure with a total height to cater to the elands' natural behavior of hitting the toys with their horns. This height ensures that the toys are accessible to the elands without posing a risk of entanglement. Testing: Measure the height from the fence to the horizontal beam to confirm that the gap is 2 feet or more. Verify with a tape measure to ensure accuracy and consistency.

### **2. Strength and Durability (High Priority)**

Given that Loki, the largest male eland, weighs approximately 1300 pounds, the hangers must support toys weighing up to 100 pounds without breaking or bending. This ensures the hangers can withstand the force exerted by the elands during play. Testing: Conduct stress tests by applying weight more than 100 pounds to the hangers and observing for any signs of bending, breaking, or structural weakness over time. Ensure the hangers remain intact and functional.

### **3. Safety (High Priority)**

Safety is crucial to prevent entanglement and injuries for both the animals and the zookeepers. The hangers must be designed to be operated by a single person from outside the pen, minimizing direct contact with the animals. Testing: Conduct a risk assessment to ensure there are no sharp edges or points on the hangers. Additionally, simulate the operation of the hangers by a single person from outside the pen to confirm ease of use and safety.

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#### **4. Accessibility (High Priority)**

The hangers should be easy to operate by a single person from outside the pen, incorporating a crane-like structure with a crank. This ensures that the zookeepers can manage the enrichment toys without entering the enclosure. Testing: Have multiple zookeepers test the operation of the hangers from outside the pen. Observe and record the ease of use, the time taken, and any difficulties encountered to ensure accessibility.

#### **5. Longevity (Medium Priority)**

The hangers need to be made of durable materials to ensure long-lasting functionality, reducing the need for frequent replacements. Testing: Monitor the hangers over an extended period, checking for signs of wear and tear, rust, or degradation. Conduct regular maintenance checks to ensure longevity and durability.

#### **6. Cost-effectiveness (Medium Priority)**

Adhering to the budget of \$200 is important to ensure the project remains financially feasible while still meeting the requirements. Testing: Conduct a cost analysis of the materials and construction process. Ensure that the total cost does not exceed the allocated budget of \$200 while still maintaining quality and functionality.

#### **7. Environmentally Friendly to Elands (Very Low Priority)**

The design must not negatively impact the Eland, the product has to be made out of environmentally friendly resources from its original habitat, or must be replicated/ to look like it is from the eland's original habitat. Testing: Consult with wildlife experts or conservationists to verify that the design elements, such as the tree-like structure and

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savanna-inspired details, do not negatively impact elands or their habitat if used in conservation areas.

Digital Signature of Primary Stakeholder (Zookeeper):